Research activity

A curve close to a line — 30 min.

Consider the function f defined on $\mathbb{R} \setminus \{1\}$ by $f(x) = \frac{3x-2}{x-1}$. Let $\mathcal{C}_f$ be its curve in a plane frame of reference.

1. Show that, for every $x \neq 1$, $f(x) = 3 + \frac{1}{x-1}$.
2. Determine the limits of f at $-\infty$, at 1^-, at 1^+ and at $+\infty$.
3. Study the sign of $f(x) - 3$, then state the position of $\mathcal{C}_f$ relative to the line with equation $y = 3$.
4. Determine a real number $A > 1$ such that, for every $x > A$, $|f(x) - 3| < 0.01$.
5. Does the curve $\mathcal{C}_f$ cross its horizontal asymptote? Justify, and explain why a graphing window that is too wide can give a misleading impression.